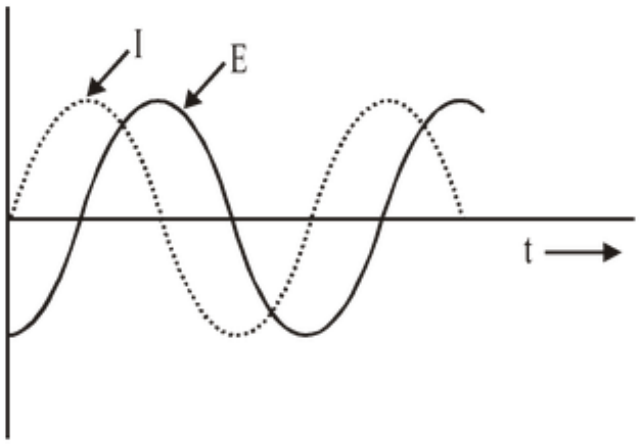
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

When an AC source of e.m.f. $E = E_0 \sin(100t)$ is connected across a circuit, the phase difference between the e.m.f. E and the current I in the circuit is observed to be $\pi/4$, as shown in the figure. If the circuit consists possibly only of $\mathrm{R}-\mathrm{C}$ or $\mathrm{R}-\mathrm{L}$ or $\mathrm{L}-\mathrm{C}$ in series, what will be the relation between the two elements of the circuit?



(A) $\mathrm{R} = 1\mathrm{k}\Omega, \mathrm{C} = 10\mu\mathrm{F}$

(B) $R = 1\mathrm{k}\Omega, \mathrm{C} = 1\mu\mathrm{F}$

(C) $R = 1\mathrm{k}\Omega, \mathrm{L} = 10\mathrm{H}$

(D) $\mathrm{R} = 1\mathrm{k}\Omega, \mathrm{L} = 1\mathrm{H}$

Q.2

A short-circuited coil is placed in a time-varying magnetic field. Electrical power is dissipated due to the current induced in the coil. If the number of turns were to be quadrupled and the wire radius halved, the electrical power dissipated would be

(A) halved

(B) the same

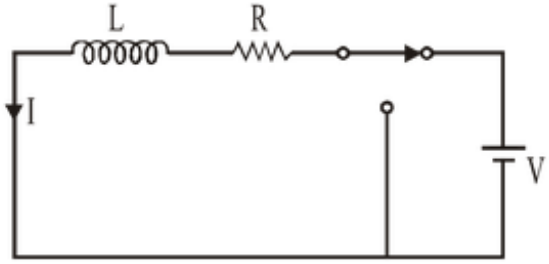
(C) doubled

(D) quadrupled

Q.3

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In the circuit shown, how soon will the coil current reach η fraction of the steady-state value?



(constant voltage source)

(A) $\frac{\mathrm{L}}{\mathrm{R}}$

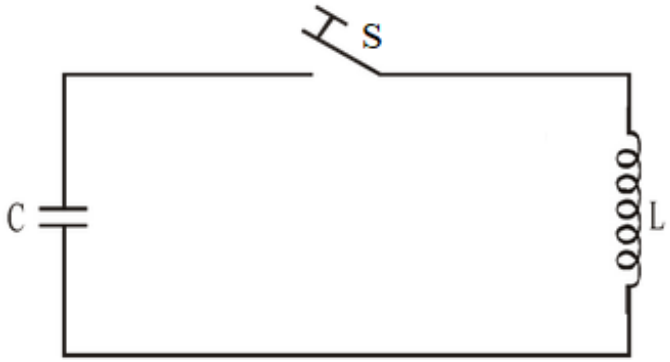
(B) $\frac{\mathrm{L}}{\mathrm{R}} \ln \frac{\eta}{(1-\mathrm{n})}$

(C) $\frac{\mathrm{L}}{\mathrm{R}} \ln \frac{1}{(1-\eta)}$

(D) $\frac{\mathrm{L}}{\mathrm{R}} \ln(1-\eta)$

Q.4

The capacitor shown in the figure is initially charged to a voltage V and the circuit is then closed. The charge of C is found to oscillate with frequency ω. the frequency will be doubled if



(A) the voltage V is doubled

(B) both L and C are halved

(C) both L and C are doubled

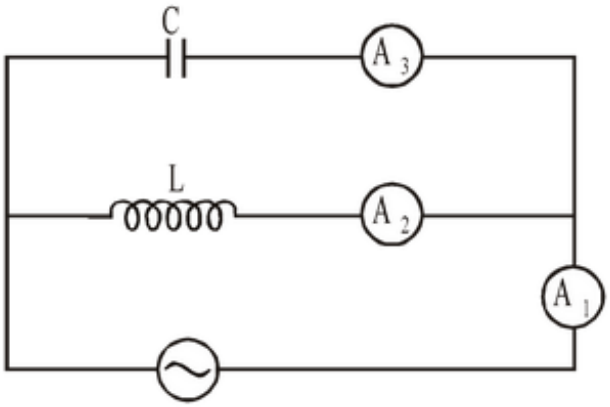
(D) both L and C are increased by a factor $\sqrt{2}$

Q.5

An inductor L, a capacitor C and ac ammeters A_1, A_2 and A_3 are connected in the circuit as shown. When the

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frequency of the oscillator is increased than at resonant frequency, the reading of ammeter



(A) A_1 is zero

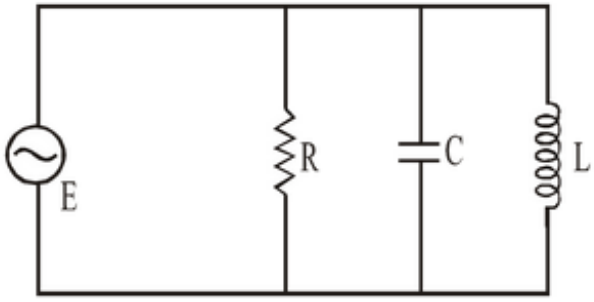
(B) A_2 is zero

(C) A_3 is zero

(D) none of these

Q.6

In a parallel LCR circuit shown in the fig. at resonance



(A) the source current is maximum

(B) the impedance of the circuit is minimum and is equal to R

(C) the resonance frequency will be the same as for a series resonance circuit with the same values of L, C and R

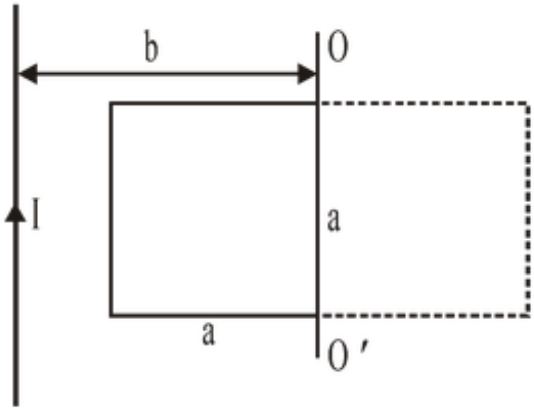
(D) the voltage across L and C are in phase

Q.7

A square frame with side ' a' and a straight conductor carrying current I are located in the same plane. The inductance and the resistance of the frame are equal to L and R respectively. The frame was turned through 180° about the axis OO' separated from the current carrying conductor by a distance b. What is the electric charge flown

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through the frame?



(A) $\frac{\mu_0 I}{AR}\ln\left(\frac{b+a}{b-a}\right)$

(B) $\frac{\mu_0 aI}{2\pi R}\ln\left(\frac{b+a}{b-a}\right)$

(C) $\frac{2\mu_0 aI}{\pi R}\ln\left(\frac{b+a}{b-a}\right)$

(D) $\frac{\mu_0 aI}{2\pi R}\ln\left(\frac{b-a}{b+a}\right)$

Q.8

An alternating current is given by the following equation: $I = 3\sqrt{2}\sin(100\pi t + \pi/4)$. Give the frequency of the current.

(A) 50

(B) 100

(C) 25

(D) 125

Q.9

When $100\ \text{Vdc}$ is applied across a solenoid, a current of $1\ \text{A}$ flows in it. When $100\ \text{V}$, ac is applied across the same coil, the current drops to $0.5\ \text{A}$. The frequency of the ac is $50\ \text{Hz}$. The impedance and inductance of the solenoid are :

(A) $100\Omega, 0.75\text{H}$

(B) $100\Omega, 0.60\text{H}$

(C) $200\Omega, 0.55\text{H}$

(D) $200\Omega, 0.75\text{H}$

Q.10

A resistance of 300Ω and an inductance of $\frac{1}{\pi}$ henry are connected in series to a ac voltage of 20 volts and $200\ \text{Hz}$ frequency. The phase angle between the voltage and current is

(A) $\tan^{-1}\frac{4}{3}$

(B) $\tan^{-1}\frac{3}{4}$

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(C) $\tan^{-1}\frac{3}{2}$

(D) $\tan^{-1}\frac{2}{5}$

Q.11

Alternating current cannot be measured by dc ammeter because

(A) ac cannot pass through dc ammeter

(B) Average value of complete cycle is zero

(C) ac is virtual

(D) ac changes its direction

Q.12

Power delivered by the source of the circuit becomes maximum, when

(A) $\omega L = \omega C$

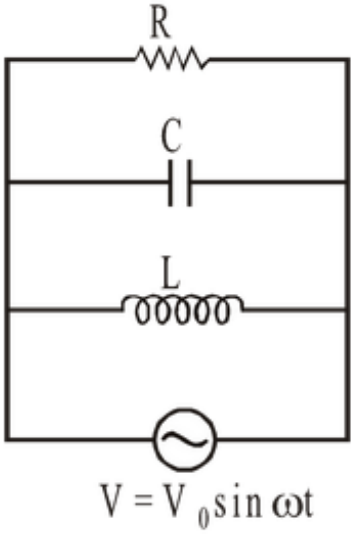
(B) $\omega L = \frac{1}{\omega C}$

(C) $\omega L = -\left(\frac{1}{\omega C}\right)^2$

(D) $\omega L = \sqrt{\omega C}$

Q.13

In the circuit shown in figure $X_L = \frac{X_c}{2} = R$, the peak value current i_0 is



(A) $\frac{\sqrt{5}V_0}{2R}$

(B) $\frac{V_0}{2\sqrt{2R}}$

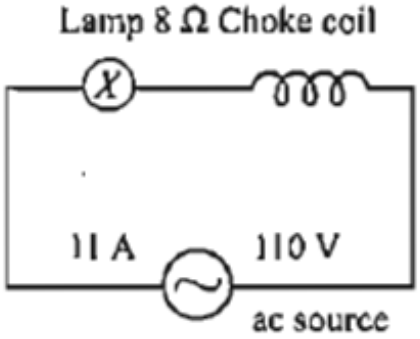
(C) $\frac{V_0}{2R}$

(D) $\frac{V_0}{2\sqrt{3R}}$

Q.14

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A lamp with a resistance of $8\ \mathrm{W}$ is connected to a choke coil as shown in Fig. 10.17. This arrangement is connected to an alternating source of $110\ \mathrm{V}$. The current in the circuit is $11\ \mathrm{A}$. The frequency of the ac is $60\ \mathrm{Hz}$. Find the value of inductive reactance of the choke coil.



(A) 6Ω

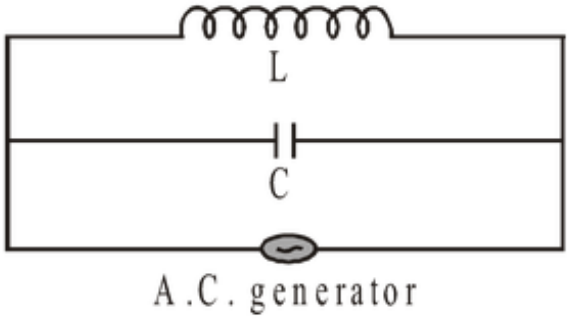
(B) 5Ω

(C) 7Ω

(D) 2Ω

Q.15

In the alternating current shown in the figure, the currents through the inductor and capacitor are $1.2\mathrm{amp}$ and $1.0\mathrm{amp}$ respectively. The current drawn from the generator is



(A) $0.4\mathrm{amp}$

(B) $0.2\mathrm{amp}$

(C) $1.0\mathrm{amp}$

(D) $1.2\mathrm{amp}$

Q.16

The phase difference between the ac and e.m.f. is $\pi/2$. Which of the following cannot be the constituent of the circuit

(A) LC

(B) L alone

(C) C alone

(D) $\mathrm{R, L}$

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Q.17

In a series resonant LCR circuit, the voltage across R is 100 volts and $\mathrm{R} = 1K\Omega$ with $\mathrm{C} = 2\mu\mathrm{F}$. The resonant frequency ω is $200\mathrm{rad/s}$. At resonance, the voltage across L is:

(A) $40\ \mathrm{V}$

(B) $250\ \mathrm{V}$

(C) $4 \times 10^{-3}\ \mathrm{V}$

(D) $2.5 \times 10^{-2}\ \mathrm{V}$

Q.18

In a series LCR the voltage across resistance, capacitance and inductance is $10\ \mathrm{V}$ each. If the capacitance is short circuited, the voltage across the inductance will be

(A) $\frac{10}{\sqrt{2}}\ \mathrm{V}$

(B) $10\ \mathrm{V}$

(C) $10\sqrt{2}\ \mathrm{V}$

(D) $20\ \mathrm{V}$

Q.19

A coil, a capacitor and an AC source of voltage $24\ \mathrm{V(rms)}$ are connected in series. By varying the frequency of the source, a maximum rms current of $6\ \mathrm{A}$ is observed. If this coil is connected to a battery of e.m.f. $12\ \mathrm{V}$ and internal resistance 4Ω, the current through it will be :

(A) $2.0\ \mathrm{A}$

(B) $1.5\ \mathrm{A}$

(C) $3.0\ \mathrm{A}$

(D) $2.5\ \mathrm{A}$

Q.20

In a series R-L-C AC circuit, for a particular value of R, L and C, power supplied by the source is P at resonance. If the value of inductance is halved, then the power from the source again at resonance is P'. Then

(A) $P = \frac{P'}{2}$

(B) $P = 2P'$

(C) $P = 4P'$

(D) $P = P'$

Q.21

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A circuit consists of a capacitor and a resistor having resistance $R = 220\ \text{ohm}$ connected in series. When an alternating e.m.f. of peak voltage $V = V_0 = 220\sqrt{2}$ is applied to the circuit, the peak current in a steady state is observed to be $I = 1\ \text{A}$. The phase difference (in degree) between the current and the voltage is

Q.22

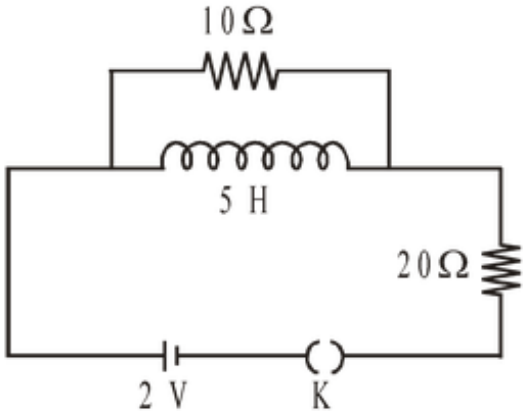
A step up transformer is used on $120\ \text{V}$ line to provide a potential difference of $2400\ \text{V}$. If the number of turns in primary is 75, then the number of turns in the secondary shall be

Q.23

An ac source of angular frequency ω is fed across a resistor R and a capacitor C in series. The current registered, If the frequency of source is changed to $\omega/3$ (maintaining the same voltage), the current in the circuit is found to be halved. The ratio of reactance to resistance at the original frequency ω is $\frac{\sqrt{A}}{\sqrt{B}}$, where A and B are relatively prime. Find the value of $\text{A} + \text{B}$.

Q.24

Two resistances of 10Ω and 20Ω and an ideal inductor of inductance 5H is connected to a $2\ \text{V}$ battery through a key K, as shown in fig. The key is inserted at $\text{t} = 0$. What is the final value of current in the 10Ω resistor ?



Q.25

A motor having an armature of resistance 2Ω is designed to operate at $220\ \text{V}$ mains. At full speed, it develops a back e.m.f. of $210\ \text{V}$. What is the current in the armature when the motor is running at full speed?

Q.26

An LCR series circuit with $\text{R} = 100\Omega$ is connected to a $200\ \text{V}, 50\ \text{Hz}$ a.c. source. When only the capacitance is removed, the voltage leads the current by 60°. When only the inductance is removed, the current leads the voltage by 60°. The current in the circuit is

Q.27

An ac ammeter is used to measure current in a circuit. When a given direct current passes through the circuit, the

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ac ammeter reads 3 A. When another alternating current passes through the circuit, the ac ammeter reads 4 A. Then the reading of this ammeter, if dc and ac flow through the circuit simultaneously, is

Q.28

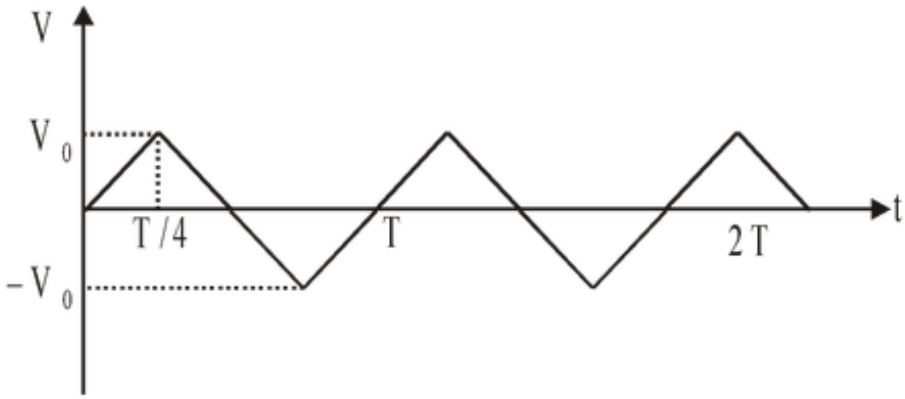
In a LCR circuit capacitance is changed from C to $2C$. For the resonant frequency to remain unchanged, the inductance should be changed from L to K. Find L/K.

Q.29

In an LCR series ac circuit, the voltage across each of the components, L, C and R is $50V$. The voltage across the LC combination will be

Q.30

The voltage time $(V-t)$ graph for a triangular wave having peak value V_0 is as shown in figure. The rms value of voltage is $\frac{Vo}{\sqrt{K}}$, where K is a positive integer. Find K.



Answers & Solutions

Q.1 Answer:

$\mathrm{R} = 1\mathrm{k}\Omega, \mathrm{C} = 10\mu\mathrm{F}$

Solution:

Given $E = E_0 \sin(100t)$. comparing this with $E = E_0 \sin\omega = 100\mathrm{rads}^{-1}$. It follows from the figure that the current leads to the e.m.f. which is true only for R-C circuits, and not for R-L circuits. Hence the circuit does not contain an inductor. For R-C circuit, the phase difference between E and I given by $\tan\phi = \frac{1}{\omega RC}$
Given $\phi = \pi/4$. Also $\omega = 100\mathrm{rads}^{-1}$. Using these values in (i), we get $\tan\left(\frac{\pi}{4}\right) = \frac{1}{100\mathrm{RC}}$ or $\mathrm{RC} = \frac{1}{100}$

Q.2 Answer:

the same

Solution:

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The magnitude of the induced voltage is proportional to the rate of change of magnetic flux which, in turn, depends on the number of turns in the coil, i.e. $\mathrm{V} \propto \mathrm{n}$ **The resistance of a wire is given by** $R = \frac{\rho l}{\pi r^2}$ **or** $R \propto \frac{l}{r^2}$**. Here** ρ **is the resistivity of the material of the wire.**

$\therefore$ Power $P = \frac{V^2}{R} \propto \frac{n^2}{l/r^2}$ or $P \propto \frac{(nr)^2}{l}$

$\therefore \frac{P_2}{P_1} \left(\frac{n_2}{n_1}\right)^2 \times \left(\frac{r_2}{r_1}\right)^2 \times \left(\frac{l_1}{l_2}\right)$

Now, if a wire of length l_1 **and radius** r_1 **is stretched to a length** l_2 **such that its radius reduced to** r_2**, then (since the mass of the wire remains constant)**

$m = \pi r_1^2 l_1 = \pi r_2^2 l_2 d$ **(d is the density)**

or $\frac{l_1}{l_2} = \left(\frac{r_2}{r_1}\right)^2$

$\frac{P_2}{P_1} = \left(\frac{n_2}{n_1}\right)^2 \times \left(\frac{r_2}{r_1}\right)^4$

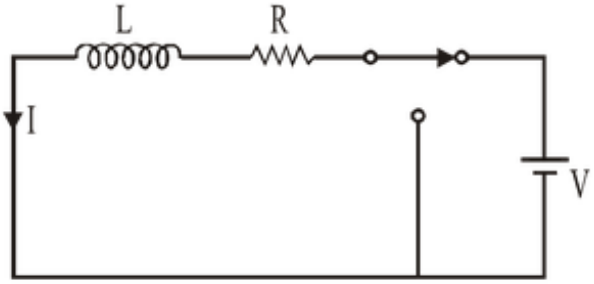
Given $\frac{n_2}{n_1} = 4$ **and** $\frac{r_2}{r_1} = \frac{1}{2}$**. Using these values, we get** $\frac{P_2}{P_1} = 4^2 \times \left(\frac{1}{2}\right)^4 = 1$

Q.3 Answer:

$\frac{\mathrm{L}}{\mathrm{R}} \ln \frac{1}{(1-\eta)}$

Solution:

From Kirchhoff's law, we get



$\mathrm{V} - \mathrm{L}\frac{\mathrm{dI}}{\mathrm{dt}} = \mathrm{IR}$ or $\frac{\mathrm{dI}}{\mathrm{I}-\frac{\mathrm{V}}{\mathrm{R}}} = -\frac{\mathrm{R}}{\mathrm{L}}\mathrm{dt}$

Upon integration, one obtains the current at any time t **:** $l(t) = \frac{V}{R}\left(1 - e^{-tR/L}\right)$

The steady value is $I_0 = \frac{V}{R}$**. Now** $\frac{I(t)}{I_0} = \eta$ **according to the problem. This gives the corresponding time** $\tau = \frac{\mathrm{L}}{\mathrm{R}} \ln \frac{1}{(1-\eta)}$

Q.4 Answer:

both L and C are halved

Solution:

$\omega' = \sqrt{\frac{1}{L/2 \cdot C/2}} = \sqrt{\frac{4}{LC}} = 2\left(\sqrt{\frac{1}{LC}}\right) = 2\omega$

Obviously, the choice (b) is correct

Q.5 Answer:

A_1 is zero

Solution:

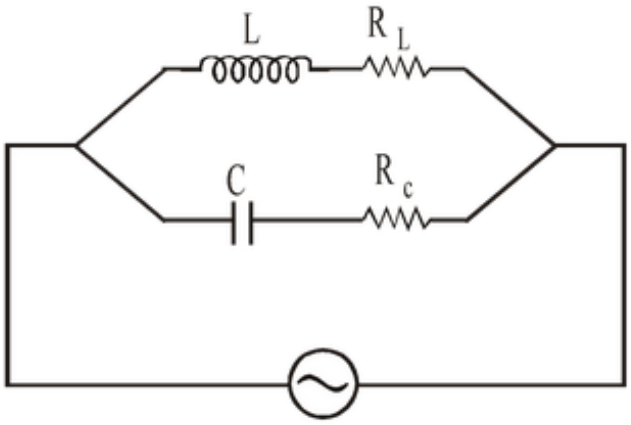
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At resonance $\omega \mathrm{L}=\frac{1}{\omega C}$ so the reading of A_1 is zero.

Q.6 Answer:

the resonance frequency will be the same as for a series resonance circuit with the same values of L, C and R

Solution:



In a parallel resonance circuit, at resonance $\omega=\frac{1}{\sqrt{\mathrm{LC}}} \sqrt{\frac{\mathrm{L} / \mathrm{C}-\mathrm{R}_{\mathrm{L}^2}}{\mathrm{L} / \mathrm{C}-\mathrm{R}_{\mathrm{C}^2}}}$

For a good quality capacitor $R_c=0$ and we put $R_L=R$. If resistance is neglected then the condition of resonance in both series and parallel circuits become the same. Namely

$\omega=(\mathrm{LC})^{-1 / 2}$

Q.7 Answer:

$\frac{\mu_0 a I}{2 \pi R} \operatorname{In}\left(\frac{b+a}{b-a}\right)$

Solution:

According to ohm's law and Faraday's law of induction, the current i_0 appearing the frame during its rotation is determined by the formula $i_0=\frac{d \phi}{d t}=-\frac{L d i_0}{d t}$ Hence, the required amount of electric charge is

$q=\int i_0 d t=-\frac{1}{R} \int\left(d \phi+L d i_0\right)$

$=\frac{1}{R} \int\left(\Delta \phi+L \Delta i_0\right)$

Where R is the resistance and L is the inductance of the frame. Since the frame has been stopped after rotation, the current in it vanishes eventually. Therefore $\Delta i_0=0$. It remains for us to find the increment of the flux $\Delta \phi$ through the frame $\left(\Delta \phi=\phi_2-\phi_1\right)$. Let us choose the normal $\hat{n}$ to the plane of the frame such that in the final position, $\hat{n}$ is directed perpendicular to the plane and towards it. Then it can easily seen that in the final position $\phi_2>0$. While in the initial position $\phi_1>0$ and $\Delta \phi$ turns out to be simply equal to the flux through the surface bounded by the final and initial positions of the frame:

$\Delta \phi=\phi_2-\left(-\phi_1\right)=\int_{b-a}^{b+a} B a d r$

Where B is a function of r, whose form can be easily found with the help of the Ampere's circuital law or Biot-Savart law. Finally, we obtain $q=\frac{\Delta \phi}{R}=\frac{\mu_0 a I}{2 \pi R} \operatorname{In}\left(\frac{b+a}{b-a}\right)$

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Q.8 Answer:

50

Solution:

$I = 3\sqrt{2}\sin(100\pi t + \pi/4)$
$I_{\text{mean}} = I_{\text{peak}}/\sqrt{2} = 3\sqrt{2}/\sqrt{2}\ \text{A} = 3\ \text{A}$
and $2\pi f = 100\pi \Rightarrow f = 100\pi/2\pi = 50\ \text{Hz}$
Frequency $(f) = 50\ \text{Hz}$

Q.9 Answer:

$200\Omega, 0.55\text{H}$

Solution:

$\text{R} = \frac{100}{1} = 100\Omega \quad \left(\text{R} = \frac{\text{V}}{\text{i}} \text{ for DC}\right) \quad ...(\text{i})$
$0.5 = \frac{100}{\sqrt{\text{R}^2 + (2\pi\text{fL})^2}} \quad ...(\text{ii})$

From Eqs. (i) and (ii), we get

$L = 0.55\text{H}$ and $\sqrt{\text{R}^2 + (2\pi\text{fL})^2} = 200\Omega$

Q.10 Answer:

$\tan^{-1}\frac{4}{3}$

Solution:

Phase angle $\tan\phi = \frac{\omega L}{R} = \frac{2\pi \times 200}{300} \times \frac{1}{\pi} = \frac{4}{3}$
$\therefore \quad \phi = \tan^{-1}\frac{4}{3}$

Q.11 Answer:

Average value of complete cycle is zero

Solution:

In dc ammeter, a coil is free to rotate in the magnetic field of a fixed magnet.
If an alternating current is passed through such a coil, the torque will reverse its direction each time the current changes direction and the average value of the torque will be zero.

Q.12 Answer:

$\omega L = \frac{1}{\omega C}$

Solution:

$\angle\rho = \text{I}^2\text{R}$

Q.13 Answer:

$\frac{\sqrt{5}V_0}{2R}$

Solution:

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$\frac{1}{Z}=\frac{1}{R}+\frac{1}{J(2R)}-\frac{1}{J(R)}=\frac{1}{R}-\frac{1}{2RJ}=\frac{2RJ-R}{2R^2J}=\frac{1}{2R}(J+2)$
$\bar{Z}=\frac{2R}{J+2}=\frac{2R(J-2)}{-5}$
$Z=\frac{2R}{\sqrt{5}} \quad \therefore i_0=\frac{V_0}{Z}=\frac{\sqrt{5}V_0}{2R}$

Q.14 Answer:

6Ω

Solution:

$R=0.8\Omega, \omega=2\pi\times 60\ \text{Hz}, l=1\ \text{A}, V=110\ \text{V}$
$Z^2=R^2+X_L^2$
$10^2=8^2+X_L^2 \quad$ **or** $\quad X_L=\sqrt{(100-64)}=\sqrt{36}=6\Omega$

Q.15 Answer:

0.2amp

Solution:

According to ohm's law
$\mathrm{i_L}=\frac{\mathrm{V_L}}{\mathrm{X_L}} \& \mathrm{i_C}=\frac{\mathrm{V_C}}{\mathrm{X_C}}$
current from AC **generator** i_L-i_c

Q.16 Answer:

$\mathrm{R, L}$

Solution:

The phase difference between the ac and emf in RL circuit varies between zero to $\pi/2$**, but never equal to** $\pi/2$

Q.17 Answer:

$250\ \text{V}$

Solution:

In resonance condition $\mathrm{V=V_R, V_L=V_C}, \omega=\frac{1}{\sqrt{LC}}$
$\therefore V_L=V_C$ i.e., $iX_C=\frac{i}{\omega C}=\frac{V}{\omega CR}$
$=\frac{100}{200\times 2\times 10^{-6}\times 10^3}$
$\Rightarrow V_L=250\ \text{V}$

Q.18 Answer:

$\frac{10}{\sqrt{2}}\ \text{V}$

Solution:

Since, $\mathrm{V_L=V_C=V_R}$
$\therefore \quad X_L=X_C=R$ **and** $V=V_R=10V$

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when capacitor is short circuited

$i = \frac{10}{\sqrt{R^2+X_L^2}} = \frac{10}{\sqrt{2R}}$

$V_L = iX_L = \left(\frac{10}{\sqrt{2R}}\right) \mathrm{R} = \frac{10}{\sqrt{2}}\ \mathrm{V}$

Q.19 Answer:

$1.5\ \mathrm{A}$

Solution:

With an AC source, current in the circuit is maximum when $\mathrm{Z} = \mathrm{Z}_{\min} = \mathrm{R}$ **(the resistance of coil)**

$\therefore \mathrm{R} = \frac{24}{6} = 4\Omega$

When connected with an dc source of e.m.f. $12\ \mathrm{V}$.

$\mathrm{i} = \frac{12}{\mathrm{r}+\mathrm{R}} \quad (\mathrm{r} = \text{ internal resistance of source })$

$= \frac{12}{4+4} = 1.5\ \mathrm{A}$

Q.20 Answer:

$P = P'$

Solution:

Since at resonance $P = \frac{V_{rms}^2}{R}$**. It doesn't depends on** $L \Rightarrow P' = P$

Q.21 Answer:

—

Solution:

$1 = \frac{220\sqrt{2}}{\sqrt{R^2+X_c^2}}$

$R^2 + X_C^2 = 220 \times 220 \times 2$

$X_c^2 = \left[2 \times (220)^2\right] - [220]^2$

$X_C = (220)$

$\tan\phi = \frac{Xc}{R} = \frac{220}{220} = 1$

$\phi = \mathbf{45}^\circ$

Q.22 Answer:

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Solution:

Here $E_p = 120\ \mathrm{V}, \mathrm{E_s} = 2400\ \mathrm{V}, \mathrm{np} = 75$ **and** $\mathrm{ns} = ?$

$\text{As } \frac{\mathrm{E_S}}{\mathrm{E_P}} = \frac{\mathrm{ns}}{\mathrm{n_p}} \Rightarrow \frac{2400}{120} = \frac{\mathrm{ns}}{75}$

$\Rightarrow \mathrm{ns} = \frac{2400}{120} \times 75 = 1500$

Q.23 Answer:

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Solution:

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According to the given problem

$I = \frac{V}{Z} = \frac{V}{\left[R^2 + \frac{1}{(\omega C)^2}\right]^{\frac{1}{2}}}$ **...(i)**

$\frac{1}{2} = \frac{V}{\left[R^2 + \frac{3}{(\omega C)^2}\right]^{\frac{1}{2}}}$ **...(ii)**

From (i) and (ii)

$\left(R^2 + \frac{1}{\omega^2 C^2}\right) = R^2 + \frac{9}{\omega^2 C^2}$

or $\frac{1}{\omega^2 C^2} = \frac{3}{5} R^2$

$\therefore \quad \frac{X}{R} = \frac{\left(\frac{1}{\omega C}\right)}{R} = \frac{\left(\frac{3}{5} R^2\right)^{\frac{1}{2}}}{R} = \sqrt{\frac{3}{5}}$

Q.24 Answer:

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Solution:

Since the resistance of an ideal inductor is zero, the final value of the current in the 10 Ω resistance is zero.

Q.25 Answer:

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Solution:

$\mathrm{R_a} = 2\Omega, \mathrm{E_a} = 220\ \mathrm{V}, \mathrm{E_b} = 210\ \mathrm{V}$. **Current at full speed is**

$I_a = \frac{E_a - E_b}{R_a} = \frac{220 - 210}{2} = 5A$

Q.26 Answer:

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Solution:

When capacitance is removed, the circuit contains only inductance and resistance. Phase difference θ between the current and voltage is then given by

$\tan\theta = \frac{\omega L}{R}$ or $\omega L = R\tan\theta = 100\tan 60^\circ$

When the circuit contains only capacitance and resistance, the phase difference between the voltage and current is given by $\tan\phi = \frac{1}{RC\omega}$

$\therefore \quad \frac{1}{C\omega} = R\tan\phi = 100\tan 60^\circ$

The impedance of the LCR circuit is given by

$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{C\omega}\right)^2} = \sqrt{R^2 + \left(100\tan 60^\circ - 100\tan 60^\circ\right)^2} = R = 100\Omega$

The current is given by

$\mathrm{I} = \frac{\mathrm{V}}{\mathrm{R}} = \frac{200}{100} = 2\ \mathrm{A}$

Q.27 Answer:

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Solution:

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Quantity of heat liberated in the ammeter of resistance R
(i) due to direct current of $3\ \mathrm{A} = \left[(3)^2\mathrm{R/J})\right]$
(ii) due to alternating current of $4\ \mathrm{A} = \left[(4)^2\mathrm{R/J}\right] = \frac{(3)^2\mathrm{R}}{\mathrm{J}} + \frac{(4)^2\mathrm{R}}{\mathrm{J}} = \frac{25\mathrm{R}}{\mathrm{J}}$ **Let the equivalent alternating current be I virtual A, then**
$\frac{\mathrm{I}^2\mathrm{R}}{\mathrm{J}} = \frac{25\mathrm{R}}{\mathrm{J}} \quad \text{or} \quad \mathrm{I} = 5\ \mathrm{A}$

Q.28 Answer:

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Solution:

$v_0 \frac{1}{2\pi}\sqrt{LC}$
If C **changes to** $2C$ **then for keeping** v_0 **constant** L **must change to** $L/2$**.**

Q.29 Answer:

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Solution:

Net voltage across LC combination
$= V_L - V_c = 0V$

Q.30 Answer:

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Solution:

$V(t) = \left(\frac{4V_0}{T}\right)t \quad$ **for** $0 \le t \le \frac{T}{4}$
$\therefore \quad \langle V^2 \rangle_{0-T/4} = \frac{\left(\frac{16V_0^2}{T^2}\right)\int_0^{T/4} t^2 dt}{\int_0^{T/4} dt} = \frac{V_0^2}{3}$
$V^2 = \left(\frac{16V^2}{T^2}\right)t^2$
$\therefore \quad V_{rms} = \sqrt{\frac{V_0^2}{3}} = \frac{V_0}{\sqrt{3}}$

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